StatTesting Instructions

Last Updated: 10/20/2026 by Clara Shin (clara.shin@wsu.edu)

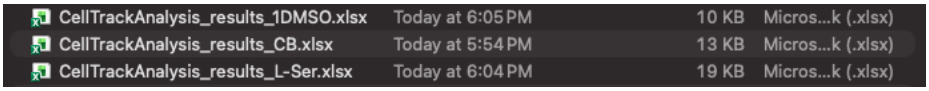
This is the **fourth step** of chemotaxis analysis in TumbleWeed. `StatTesting.m` performs statistical testing across multiple experimental groups. It provides 4 types of testing and uses statistical testing packages of MATLAB.

1. Two-Sample Kolmogorov-Smirnov (KS) test (kstest2)
2. Kruskal-Wallis test (kruskalwallis)
3. Two-Sample Student's t-test (ttest2)
4. One-way ANOVA (anova1)

- Input: `CellTrackAnalysis_results.xlsx` files from each experimental group (Output file from Step 2, `CellTrackAnalysis.m`)
- Output
    - Printout of statistical testing results.

Step 0: Prepare Excel Files

1. Important: Every time after running Step 2, change the name of excel output file (`CellTrackAnalysis_results.xlsx`). For example, add the experimental group names after each run like below:



Step 1: StatTesting Parameters

1. Open MATLAB, click "Browse For Folder" icon to navigate to TumbleWeed folder
2. Find and open `StatTesting.m`
3. Adjust parameters. See the table below for parameter instructions.

Parameter Name	Value Range	Data Type	Description
`test_type`	1, 2, 3, 4	Integer	**When `test_type` = 1:** Performs two-sample KS test (kstest2) **When `test_type` = 2:** Performs Kruskal-Wallis test (kruskalwallis) **When `test_type` = 3:** Performs two-sample t-test (ttest2) **When `test_type` = 4:** Performs one-way anova (anova1)
`group_count`	> 0	Integer	Number of experimental groups
`excel_column_name`		String	Column to read from the PerTrackMetrics sheet. Examples are TumbleCount, Tumbles_per_s, MeanRunDuration_s, etc.
`pooled`	true or false	Boolean	**When pooled = false:** 1 Excel file per experimental group **When pooled = true:** Multiple Excel files per experimental group

Step 2: Interpret Results

- Example of KS Test

```
>> StatTesting
Test selected : Two-sample KS Test (kstest2)
Please select 2 Excel file(s), one per group.
  Group 1 → CellTrackAnalysis_results_CB.xlsx
  Group 2 → CellTrackAnalysis_results_L-Ser.xlsx
  Group 1: 131 valid values read.
  Group 2: 260 valid values read.

Data matrix built: 260 rows x 2 groups.

--- Two-sample KS Test (kstest2) ---
Column tested : Tumbles_per_s
Groups        : Group1, Group2

KS statistic  : 0.402466
p-value       : 0.000000
Reject H0     : true
Significance  : *** (p < 0.001)

--- Summary Statistics: Tumbles_per_s ---
Group              N        Mean      Median         Std         IQR
-----------------------------------------------------------------------
Group1           131      0.6297      0.5714      0.5864      0.9091
Group2           260      0.2022      0.0000      0.3256      0.3922
```

- Example of ANOVA

```
>> StatTesting
Test selected : One-way ANOVA (anova1)
Please select 3 Excel file(s), one per group.
  Group 1 → CellTrackAnalysis_results_CB.xlsx
  Group 2 → CellTrackAnalysis_results_L-Ser.xlsx
  Group 3 → CellTrackAnalysis_results_1DMSO.xlsx
  Group 1: 131 valid values read.
  Group 2: 260 valid values read.
  Group 3: 84 valid values read.

Data matrix built: 260 rows x 3 groups.

--- One-way ANOVA (anova1) ---
Column tested : Tumbles_per_s
Groups        : Group1, Group2, Group3

p-value       : 0.000000

--- Post-hoc Multiple Comparisons (Tukey-Kramer) ---
Significance  : *** (p < 0.001)

--- Summary Statistics: Tumbles_per_s ---
Group              N        Mean      Median         Std         IQR
-------------------------------------------------------------------
Group1           131      0.6297      0.5714      0.5864      0.9091
Group2           260      0.2022      0.0000      0.3256      0.3922
Group3            84      0.8525      0.8422      0.6267      0.8523
```

References

- Massey Jr., F. J. The Kolmogorov-Smirnov Test for Goodness of Fit. *J. Am. Stat. Assoc.* **46**, 68–78 (1951).
- kstest2 - Two-sample Kolmogorov-Smirnov test - MATLAB. https://www.mathworks.com/help/stats/kstest2.html.
- Kruskal, W. H. & Wallis, W. A. Use of Ranks in One-Criterion Variance Analysis. *J. Am. Stat. Assoc.* **47**, 583–621 (1952).
- kruskalwallis - Kruskal-Wallis test - MATLAB. https://www.mathworks.com/help/stats/kruskalwallis.html.
- Student. The Probable Error of a Mean. *Biometrika* **6**, 1–25 (1908).
- ttest2 - Two-sample t-test - MATLAB. https://www.mathworks.com/help/stats/ttest2.html.
- Fisher, R. A. *Statistical Methods for Research Workers*. (Edinburgh, London, Oliver and Boyd, 1928).
- anova1 - One-way analysis of variance - MATLAB. https://www.mathworks.com/help/stats/anova1.html.